Written resit examination paper for MC104 – AI for Cybersecurity
Kristiania University College
Spring 2025

Examination paper released: 24/1/2025 10:00

Examination deadline: 14/2/2025 10:00

Academic contact during examination: Ayan Chatterjee, ayan@nilu.no
Course contact during examination: Andrii Shalaginov, andrii.shalaginov@kristiania.no
Technical contact during examination: eksamen@kristiania.no

Exam format: individual written home examination

Final report format: use font 12 with at least 1.0 spacing. The limit is 10-15 pages max that includes abstract, report, list of bibliography in the end, figures, and tables.

Grading scale: The Norwegian grading system uses the graded scale A - F, where A is the best grade, E is the lowest pass grade, and F is fail

Weighting: 100% of the overall grade

Support materials: All support materials are allowed

Plagiarism control: We expect your own independent work. Please, use citations and quotations if there is material you want to include in the report.

Learning Outcomes

Knowledge

The student..

Is able to demonstrate the applicability and necessity from usage of Artificial Intelligence in multiple domains of cybersecurity

Is able to formulate the best data processing strategy and application of the most relevant AI model depending on the task

Skills

The student ...

Is able to apply data analysis using community-accepted tools and industry standards / formats

Is able to program training and testing phases of the AI models to tackle Big Data problems

General competence

The student...

Is able to reflect on the results of AI model and derive necessary solutions

Is able to reflect on the necessity to optimize the cybersecurity-related data processing and minimize the risk from curse of dimensionality

Exam Task

The purpose of this master's assignment is to address the application of artificial intelligence (AI) technology in the field of cybersecurity, focusing on supervised and unsupervised learning methods. The assignment addresses three tasks, each designed to examine a different aspect of the contribution of artificial intelligence to improving cybersecurity measures.

Write an authetic report in an uniform way with proper screenshots and share the codebase with datasets and executed ipynb files with assignment submission.

Task A (20% of the grade)
1. Classify and analyze different types of cyber attacks, ranging from malware and phishing to advanced persistent threats (APTs).

2. Provide an overview of how AI is employed to enhance cybersecurity, with a focus on its applications and benefits.

3. Investigate the challenges and opportunities associated with integrating AI into cybersecurity frameworks.

4. Define and explore the principles of ethical AI, emphasizing fairness, transparency, and accountability in AI systems.

Task B (10% of the grade)
1. Analyze the dataset https://www.kaggle.com/datasets/teamincribo/cyber-securityattacks/data and show how data visualization techniques help to analyze the dataset.

Task C (70% of the grade)

Select your preferred data:

Installation: Install Anaconda environment for AI coding in Jupyter notebook or set-up the Colab environment.

Datasets: Selection of appropriate datasets of your choice from below sourses for **Task C**
- https://github.com/shramos/Awesome-Cybersecurity-Datasets
- https://archive.ics.uci.edu/datasets
- https://www.kaggle.com/datasets?search=security
- https://www.kaggle.com/datasets/kartik2112/fraud-detection/data
- https://www.kaggle.com/datasets/shashwatwork/phishing-dataset-for-machine-learning/data
- https://github.com/jmnwong/NSL-KDD-Dataset
- https://github.com/gfek/Real-CyberSecurity-Datasets
- https://csr.lanl.gov/data/

Mandatory things:
- Pipeline design
- Data preparation
- Feature Engineering
- Hyperparameter tuning
- Cross-validation
- Appropriate metric selection
- Compare with multiple models with plots
- A Concreete summary as a reflection of overall analysis

- Any one of the problem (1-4) below use multi-class classification and for others you can use binary classification.

1. Classification of Malicious Activity (15%)
• Objective: Implement machine learning pipeline to classify different types of malicious activity within network traffic.
• Tasks: Select and preprocess labeled datasets, design and train a classification model, evaluate model performance, and analyze the model's decisions.

2. Anomaly Detection in Network Traffic (15%)
• Objective: Apply machine learning techniques for anomaly detection in network traffic to identify potential security threats.
• Tasks: Select and preprocess labeled datasets, design and train a classification model, evaluate model performance, and analyze the model's decisions.

3. Spam Classification (15%)
• Objective: Implement machine learning approach to classify email into spam categories.
• Tasks: Select and preprocess labeled datasets, design and train a classification model, evaluate model performance, and analyze the model's decisions.

4. Phishing Detection (15%)
• Objective: Implement machine learning approach to classify data into phishing categories.
• Tasks: Select and preprocess labeled datasets, design and train a classification model, evaluate model performance, and analyze model's decisions.

5. Anomaly Detection on Generative Data (10%)
• Objective: Apply unsupervised machine learning like clustering for data groups and visualizing.
• Tasks: Generate dataset with noise, cluster them with clustering algorithms, and visualize clusters.